

1 Problem 5 Benchmarking Arabic NLP Tasks

1.1 Task Definition and Scope

Team 27 was assigned Topic 27: **Toxicity Detection** (كشف الإساءة). The required task is to classify each Arabic sentence or comment as either toxic or non-toxic, using **F1** as the main metric. The assignment also requires evaluation of ChatGPT, Gemini, ALLaM, Jais, and Fanar using the same inputs and prompt format, and it asks for a balanced mixture of Modern Standard Arabic (MSA), Classical Arabic, and Arabic dialects.

In this benchmark, toxic means that the sentence contains a direct insult, abuse, hate, threat, harassment, hostile attack, humiliation, demeaning wording, or clearly offensive language. Non-toxic means that the sentence is neutral, factual, friendly, religious/classical advice, ordinary emotion, or disagreement without a direct attack. The application domain is Arabic online moderation, where a deployed system must recognize hostile content across dialectal, MSA, and classical forms.

The work was deliberately kept binary because the assigned metric is F1 and the task row specifies 100 comments/sentences for toxicity detection. We consider the toxic class to be the positive class. Therefore:

$$F_1 = \frac{2TP}{2TP + FP + FN},$$

where TP is toxic text correctly predicted as toxic, FP is non-toxic text incorrectly predicted as toxic, and FN is toxic text incorrectly predicted as non-toxic. This follows the standard F1 definition as the harmonic mean of precision and recall.¹

1.2 Dataset Creation and Cleaning

The final dataset contains exactly 100 Arabic samples. We rebuilt the dataset after noticing that the first generated file was not sufficiently balanced and had duplicate/template-like examples. The final version is cleaner, balanced by label, balanced by Arabic variety group, and has no duplicated text.

Table 1: Final dataset summary.

Property	Value
Total samples	100
Unique text strings	100
Modern Standard Arabic samples	34
Classical Arabic samples	33
Dialect samples	33
Toxic samples	50
Non-toxic samples	50
Remaining @ mentions in text	0
Duplicate sentences	0

¹See the scikit-learn F1 definition: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html.

Table 2: Source composition of the final dataset.

Source	Count
MPOLD social-media benchmark	67
Public-domain Classical Arabic	17
Public-domain Classical Arabic satire/hija'	16

Most of the dataset comes from the Arabic Offensive Comment Detection / MPOLD social-media benchmark. MPOLD consists of Arabic comments collected from Twitter, Facebook, and YouTube, with binary Offensive/Non-Offensive majority labels and an annotation procedure based on multiple judgments.² Classical examples were taken from public-domain Classical Arabic religious, literary, and satire/hija' sources. The final file removes anonymized social-media handles such as @User.IDX, URL placeholders, duplicate rows, and very noisy social-media fragments. After removing or softening a few overly offensive words, the labels were rechecked so that the gold label still matched the final sentence.

Table 3: Examples from each variety group and class.

Variety group	Gold label	Example
MSA	non-toxic	في فترة الصغر والمراهقة يكون من الصعب تقبل تعليقات الأصدقاء حول شعرك المجعد وهنا يأتي دور الوالدين في ادخال الثقة الى نفسية طفلهما لكن مع تقدم بالسن يصبح الأمر عاديا
MSA	toxic	أجل بسبب تغريدة يسجن ويحاكم يا مرتزقة وأكاديمية التغيير غير الله عليكم
Classical	non-toxic	إنما المؤمنون إخوة فأصلحوا بين أخويكم
Classical	toxic	أتاني كلام الجاهل ابن كيغلغ يجوب حزونا بيننا وسهولا
Dialect	non-toxic	بتعمل حلقة صغيرة عشان عندي امتحان بكرة ومتضيعليش اليوم الله عليك يا فخر اليوتيوب
Dialect	toxic	عمر انا احبك لكن شف نفسك؟ بالاول حاط صورته وسخه وبعدين كلامك مو واضح

1.3 Annotation Process

Each row has two annotation columns, `annotator1` and `annotator2`, plus a final `gold_label`. The starting labels for MPOLD examples came from the benchmark majority label, and the team then manually checked the final cleaned text. Classical examples were manually labeled after reading the sentence in context as either harmless advice/literature or hostile satire/hija'. If a label became doubtful after cleaning, the sentence was reviewed again. The final submitted file has no remaining annotator disagreement.

²MPOLD GitHub repository: <https://github.com/shammur/Arabic-Offensive-Multi-Platform-SocialMedia-C>
Hugging Face mirror used for convenient access: https://huggingface.co/datasets/asas-ai/Arabic_Offensive_Comment_Detection.

Table 4: Inter-annotator agreement on the final reviewed dataset.

Agreement measure	Value
Annotator 1 toxic / non-toxic counts	50 / 50
Annotator 2 toxic / non-toxic counts	50 / 50
Simple percent agreement	100.0%
Cohen’s κ	1.000
Remaining disagreements	0

1.4 Evaluation Setup

All five models received the same 100 inputs and the same binary instruction. ChatGPT and Gemini were called through OpenRouter because paid API access was available there. ALLaM, Jais, and Fanar were run locally through Ollama. The final run took 89.7 minutes, which made the experiment time- and effort-consuming, especially because local models were slow and had to be tested carefully.

Table 5: Models and execution method.

Report name	Exact model id	Execution method
ChatGPT	openai/gpt-4o-mini	OpenRouter API
Gemini	google/gemini-2.5-flash	OpenRouter API
ALLaM	iKhalid/ALLaM:7b	Ollama local run
Jais	jwnder/jais-adaptive:7b	Ollama local run
Fanar	yasserrmd/Fanar-1-9B-Instruct	Ollama local run

The final prompt was intentionally strict because the first prompt allowed too much freedom. The system prompt defined toxicity and required the model to output exactly one lowercase English word only: **yes** or **no**. The user prompt was:

```
Is the following Arabic text toxic?
[Arabic text]
Answer only yes or no:
```

For API and local runs, the temperature was set to 0, the maximum output length was limited to 3 tokens, and the answer parser accepted only yes/no as valid. The final JSON log records one result per sample for each model.

1.4.1 Jais first-attempt failure and prompt correction

Jais caused a major problem in the first pilot run. With the earlier pseudo-JSON prompt, it frequently ignored the required output format and translated, paraphrased, repeated the sentence, or generated unrelated text. In that pilot output, Jais had 72/100 malformed or non-compliant outputs. This was not only a classification issue; it was a prompt-following failure. After switching to the strict yes/no prompt, Jais produced valid yes/no outputs for all 100 final samples, so it is kept in the final ranking. However, its first-attempt failure is important because it shows that weaker local models may need much more constrained prompts.

Table 6: Examples of Jais non-compliance in the first pilot run.

ID	Input text	Jais raw output in pilot
1	شو هيدا كذاب اللي عم يحكي	هو كاذب من يقول هذا
2	وش هذا تافه اللي يكلم	وش هذا تافه اللي يكلم
3	الشغل وايد تعبنا	شغل العمل متعب
4	عامل إيه؟ من زمان عنك	مرحبا يا عزيزي كيف حالك؟

1.5 Quantitative Results

Table 7 gives the final quantitative results over the full 100-case dataset. Gemini achieved the highest F1 score and accuracy. ChatGPT was second by F1, followed by ALLaM and Jais. Fanar was the most conservative model: it predicted non-toxic for 75 samples, which produced high precision but low recall.

Table 7: Overall final results for Arabic toxicity detection. Toxic is the positive class.

Model	Valid	Invalid	TP	TN	FP	FN	Precision	Recall	F1	Accuracy
ChatGPT	100	0	36	49	1	14	0.973	0.720	0.828	0.850
Gemini	100	0	40	50	0	10	1.000	0.800	0.889	0.900
ALLaM	100	0	37	45	5	13	0.881	0.740	0.804	0.820
Jais	100	0	40	38	12	10	0.769	0.800	0.784	0.780
Fanar	100	0	24	49	1	26	0.960	0.480	0.640	0.730

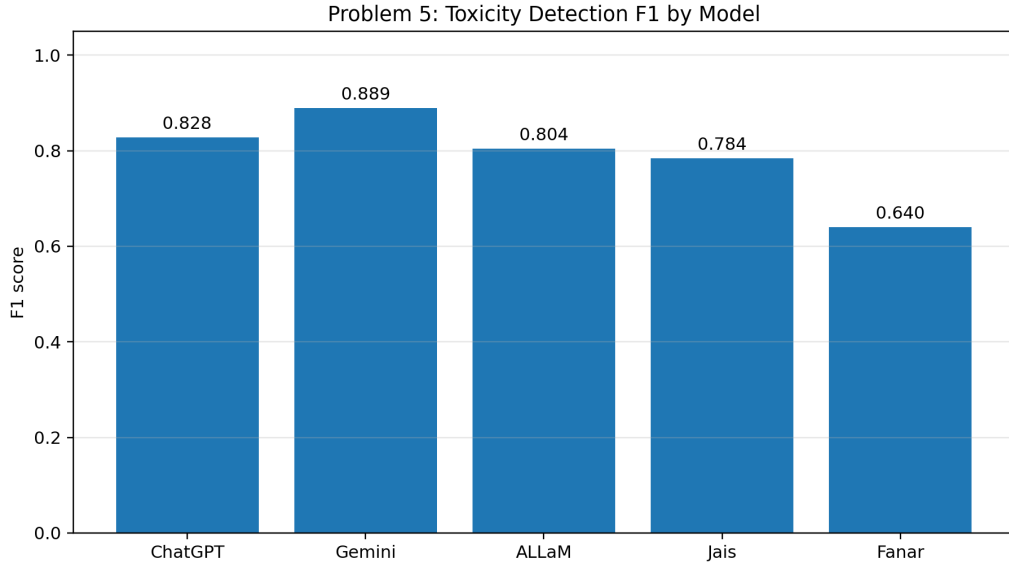


Figure 1: F1 score comparison across the five evaluated LLMs.

Table 8: F1 and accuracy by Arabic variety group.

Model	MSA F1	MSA Acc.	Classical F1	Classical Acc.	Dialect F1	Dialect Acc.
ChatGPT	0.875	0.882	0.815	0.848	0.786	0.818
Gemini	1.000	1.000	0.769	0.818	0.867	0.879
ALLaM	0.914	0.912	0.667	0.758	0.788	0.788
Jais	0.789	0.765	0.769	0.818	0.789	0.758
Fanar	0.759	0.794	0.316	0.606	0.741	0.788

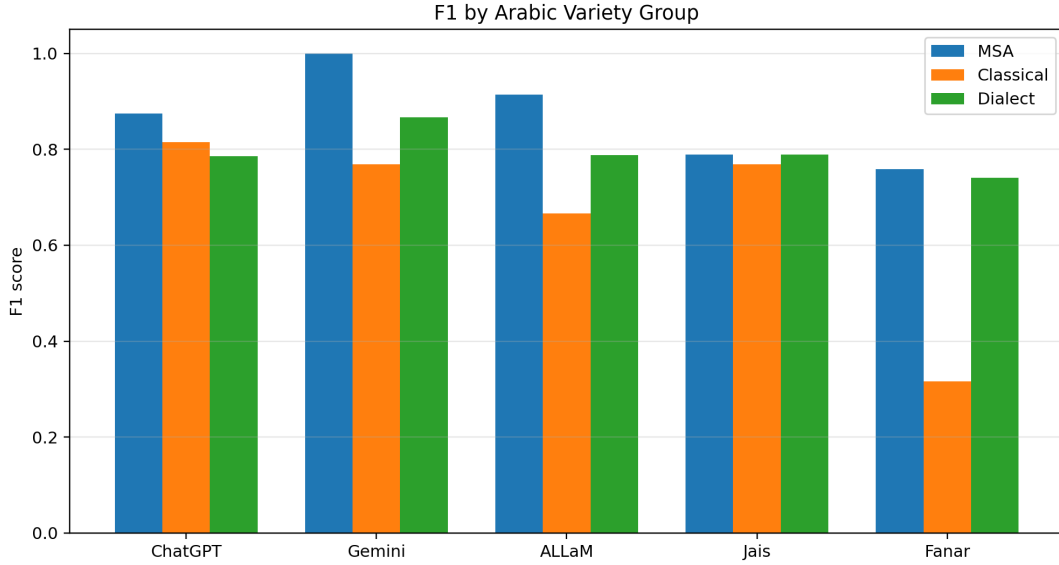


Figure 2: F1 score by Arabic variety group.

1.6 Selected Output Examples

Table 9 shows representative predictions from the final JSON log. The examples include easy cases, false positives, false negatives, and cases where several models disagreed.

Table 9: Selected model outputs from the final run.

ID	Gold	ChatGPT	Gemini	ALLaM	Jais	Fanar	Wrong models
4	non-toxic	no	no	no	no	no	none
18	toxic	yes	yes	yes	yes	yes	none
5	non-toxic	yes	no	yes	no	yes	ChatGPT, ALLaM, Fanar
11	non-toxic	no	no	yes	yes	no	ALLaM, Jais
29	toxic	no	yes	no	yes	no	ChatGPT, ALLaM, Fanar
63	toxic	no	no	no	no	no	ChatGPT, Gemini, ALLaM, Jais, Fanar
85	toxic	no	no	no	no	no	ChatGPT, Gemini, ALLaM, Jais, Fanar
93	toxic	yes	yes	no	no	no	ALLaM, Jais, Fanar

The selected outputs show that obvious toxicity is generally easy for all models, while indirect hostility, dialect slang, and Classical Arabic satire are much harder. The biggest repeated issue is false negatives: several models recognize explicit modern insults but miss softer, indirect, or older forms of attack.

1.7 Qualitative Observations

Gemini had the best overall result with $F1 = 0.889$. It made no false positives, which means it was very safe about not accusing non-toxic text. Its remaining errors were false negatives, especially in Classical Arabic and subtle dialect examples.

ChatGPT achieved $F1 = 0.828$. It had high precision (0.973) but lower recall (0.720), so it tended to miss toxic examples when the wording was indirect or literary.

ALLaM achieved $F1 = 0.804$. It was strong on MSA, where its $F1$ was 0.914, but it was weaker on Classical Arabic and produced some false positives on neutral dialect comments.

Jais achieved $F1 = 0.784$ in the final strict run. Its output format became valid, but it still produced many false positives. This means the strict prompt solved the first-attempt malfunction but did not fully solve the classification weakness.

Fanar achieved $F1 = 0.640$. Its precision was high (0.960), but its recall was only 0.480. It predicted no very often, causing many toxic samples to be missed.

1.8 Error Analysis

The assignment requires at least three incorrect or weak outputs, at least three Arabic-specific challenges, and at least two difficult/borderline cases. The following table gives representative errors from the final run.

Table 10: Representative incorrect or weak outputs.

ID	Text	Gold	Model outputs	Explanation
5	يعني مش ناقصه الوطن العربي فتنه وتشتت وانقسام بالله ترجمونا وزيادة البغضاء	non-toxic	ChatGPT=yes, Gemini=no, ALLaM=yes, Jais=no, Fanar=yes	False positive caused by negative-topic words such as fitna and hatred although the sentence asks for less division.
11	للأسف الضحايا 360 الموضوع ليس العدد، المشكلة أن النفس البشرية أصبحت أرخص شيء	non-toxic	ChatGPT=no, Gemini=no, ALLaM=yes, Jais=yes, Fanar=no	Neutral sympathy about victims; ALLaM and Jais treated sadness and death-related vocabulary as toxicity.
29	النابحين في سله واحده لذلك تجد هذه السلع الفاسده	toxic	ChatGPT=no, Gemini=yes, ALLaM=no, Jais=yes, Fanar=no	Toxic metaphor and demeaning plural wording; some models missed the insult when it appeared in indirect political language.

ID	Text	Gold	Model outputs	Explanation
51	عمر انا احبك لكن شف نفسك؟ بالاول حاط صورته وسخه وبعدين كلامك مو واضح	toxic	ChatGPT=no, Gemini=yes, ALLaM=yes, Jais=yes, Fanar=no	Dialectal Gulf sentence with mixed affection and criticism; ChatGPT and Fanar judged it as non-toxic.
63	وقطر عاملة الشريفة بقناتكم الظاهرة	toxic	ChatGPT=no, Gemini=no, ALLaM=no, Jais=no, Fanar=no	Subtle sarcastic Gulf/political attack; all five models missed it.
85	أتاني كلام الجاهل ابن كيفلغ يجوب حزونا بيننا وسهولا	toxic	ChatGPT=no, Gemini=no, ALLaM=no, Jais=no, Fanar=no	Classical satire/hija' with the word jahil; all five models treated it as harmless classical wording.
93	فاذهب فأنت طليق عرضك إنه عرض عززت به وأنت ذليل	toxic	ChatGPT=yes, Gemini=yes, ALLaM=no, Jais=no, Fanar=no	Classical insult about social status and humiliation; local models especially missed the pragmatic attack.

1.8.1 Arabic-specific challenge 1: dialect variation

Dialectal Arabic contains local words, spelling variation, and social-media shortcuts that are not always represented in model pretraining equally. Gulf examples such as **ياخوك**, Egyptian expressions, Levantine spellings, and rare Maghrebi forms require dialect awareness. The final dataset includes Gulf, Egyptian, Levantine, Iraqi, and Maghrebi items inside the dialect group. Several models missed subtle Gulf insults and sarcasm, while some local models treated harmless dialect laughter or casual statements as toxic.

1.8.2 Arabic-specific challenge 2: morphology and orthographic variation

Arabic words can carry meaning through roots, patterns, affixes, and plural forms. Toxicity may appear through a derived form rather than a direct dictionary word. For example, **النابحين** is a demeaning plural metaphor, while **الفاسده** and **الفتنة** can be toxic or non-toxic depending on context. Missing hamza, final ta marbuta spelling, and informal orthography also make exact keyword matching unreliable.

1.8.3 Arabic-specific challenge 3: ambiguity and context dependence

Some comments contain negative vocabulary but are not attacks, while others are toxic through sarcasm without using a clear slur. Sentence 5 discusses conflict and hatred but asks people to stop increasing division, so it is non-toxic. Sentence 63 is toxic mostly be-

cause of sarcastic framing and political attack. This shows that Arabic toxicity detection requires pragmatic understanding, not only word spotting.

1.8.4 Arabic-specific challenge 4: Classical Arabic satire

The Classical Arabic section was difficult because older insults and *hija'* do not look like modern social-media abuse. Several models treated hostile literary lines as neutral because the style is poetic and archaic. Fanar was especially conservative on Classical Arabic, with Classical F1 = 0.316. This suggests that modern Arabic safety classifiers may not capture older rhetorical forms of humiliation.

1.9 Difficult and Borderline Cases

Some items were difficult even for humans because they depend on social-media context, sarcasm, or literary background. Table 11 lists examples.

Table 11: Borderline or difficult cases.

ID	Text	Gold	Wrong	Why it is difficult
5	يعني مش ناقصه الوطن العربي فتنه وتشئت وانقسام بالله ترحمونا وزيادة البغضاء	non-toxic	3/5	Although it contains words related to conflict, the communicative intent is a request to reduce division, so the gold label is non-toxic.
56	احمد يعاني من مرض عضال نادر يشوش على مجاله العصبي	toxic	4/5	The final label follows the reviewed benchmark label, but the offensive signal is weak without the original social-media context.
63	وقطر عاملة الشريفة بقناكم الظاهرة	toxic	5/5	The wording is sarcastic and political rather than a direct insult; models that depend on obvious slur words miss it.
85	أتاني كلام الجاهل ابن كيغلع يجوب حزوننا وسهولا	toxic	5/5	Classical <i>hija'</i> is hard because the offensive force is literary and archaic, not modern slang.

1.10 Limitations

The benchmark is useful but still small. It contains 100 examples, so a few borderline labels can noticeably change the metric. The dataset is balanced across the three required variety groups, but dialects inside the dialect group are not equally represented because the priority was to keep the total at 100 while using real examples.

A second limitation is context. Social-media toxicity often depends on previous comments, the target person, or a named group. Our inputs were single sentences or comments, so some cases lose context. This explains why a few original MPOLD labels look less obvious after cleaning and shortening.

A third limitation is model access. ChatGPT and Gemini required paid API access, while ALLaM, Jais, and Fanar were run locally through Ollama. Local inference was slow and resource-limited. The team had to wait for long local model runs, pay for API calls, modify the prompt, rerun the benchmark, and manually inspect labels and outputs. This made the work very time- and effort-consuming. We tried our best, became tired during repeated runs and manual checking, and still documented the failures honestly.

Finally, prompt format matters. Jais failed badly in the first attempt when the required answer format was less constrained. The final strict yes/no prompt improved compliance, but a different prompt or few-shot examples could change the classification

results. Therefore, the results should be interpreted as a controlled zero-shot comparison under one strict prompt, not as a permanent ranking of the models.

1.11 Final Conclusions and Recommendations

Gemini was the best model in this benchmark, with $F1 = 0.889$ and $\text{accuracy} = 0.900$. ChatGPT came second, with very high precision but lower recall. Among local models, ALLaM was the strongest overall and performed especially well on MSA. Jais became format-compliant after the strict prompt but still made many false positives. Fanar was the most conservative model and missed many toxic examples, especially Classical Arabic attacks.

The main recommendation is to use stricter output prompts for local LLM evaluation. A simple yes/no format was much more robust than pseudo-JSON for Jais. For future work, the dataset should be expanded beyond 100 samples, dialects should be balanced inside the dialect group, and ambiguous MPOLD examples should be reviewed with more annotators. A few-shot prompt with examples of dialect sarcasm and Classical Arabic *hija*’ may also improve recall. For production moderation, a model should not rely only on binary zero-shot LLM outputs; it should combine LLM classification with annotation guidelines, confidence thresholds, and human review for borderline cases.

1.12 Checklist

Table 12: Problem 5 checklist.

Requirement	Status	Evidence in our work
Task definition	Completed	Binary Arabic toxicity detection for Team 27 / Topic 27.
Dataset size	Completed	100 samples.
Balanced Arabic varieties	Completed	34 MSA, 33 Classical, 33 Dialect.
Real data majority	Completed	MPOLD social-media comments plus public-domain Classical Arabic.
Gold output for each sample	Completed	<code>gold_label</code> column in the final CSV.
At least two annotators	Completed	<code>annotator1</code> and <code>annotator2</code> included for every sample.
IAA reported	Completed	Simple agreement = 100.0%, Cohen’s $\kappa = 1.000$.
Same prompt for all models	Completed	Strict yes/no prompt used for all five models.
All five required models	Completed	ChatGPT, Gemini, ALLaM, Jais, and Fanar evaluated.
Inputs, prompts, outputs logged	Completed	Final CSV and JSON evaluation logs are submitted with the report.
Metric reported	Completed	F1, precision, recall, accuracy, and confusion counts.
Comparison table	Completed	Table 7.

Requirement	Status	Evidence in our work
Visualization	Completed	F1 bar chart and variety-group F1 chart.
Error analysis	Completed	Representative errors, Arabic challenges, and borderline cases included.
Limitations and recommendations	Completed	Final sections discuss limitations and future improvements.

1.13 Submitted Files for Problem 5

The Problem 5 submission package should include: the final dataset CSV, the final evaluation JSON log, the evaluation script/prompt used for the final run, this `prob5.tex` report section, and the generated figures. The important files are:

- `to_annotate.csv`: final 100-sample dataset with gold labels and annotator columns.
- `evaluation.json`: final outputs for ChatGPT, Gemini, ALLaM, Jais, and Fanar.
- `prob5.tex`: final report section.
- `Figures/problem5_f1_scores.png` and `Figures/problem5_f1_by_variety.png`: report visualizations.